AEGIS BANK

Logging & Monitoring Standards

Enterprise AI Decision & Knowledge Platform for Modern Banking

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Document Information

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Confidentiality Notice

This document defines the logging, monitoring, alerting, and observability standards used by Aegis Bank.

The purpose of this standard is to ensure system reliability, security visibility, operational transparency, and rapid incident response.

Unauthorized access or modification of monitoring information is prohibited.

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1. Purpose

The purpose of this document is to establish standards for collecting, storing, analyzing, and monitoring system information across Aegis Bank infrastructure.

These standards ensure:

- System availability.
- Faster troubleshooting.
- Security threat detection.

- Compliance visibility.
- Performance optimization.

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2. Scope

This standard applies to:

- IT Operations.
- DevOps teams.
- Security teams.
- Developers.
- Database administrators.
- AI engineering teams.

It covers monitoring of:

- Banking applications.
- APIs.
- Databases.
- Cloud infrastructure.
- Docker containers.
- AI systems.
- Network services.

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3. Monitoring Principles

Aegis Bank follows these monitoring principles:

Visibility

All critical systems must provide operational visibility.

Proactive Detection

Issues should be detected before customer impact.

Security Awareness

Suspicious activities must be identified quickly.

Traceability

All important activities must have audit records.

Continuous Improvement

Monitoring data is used to improve systems.

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4. Logging Architecture

Aegis Bank follows centralized logging architecture.

Applications

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API Services

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Log Collection Layer

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Central Log Platform

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Analysis & Alerting

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Security / Operations Teams

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5. Application Logging Standards

Applications must generate structured logs.

Required information:

- Timestamp.
- Service name.
- Request ID.
- User activity.
- Error details.
- System status.

Log Levels

Level	Purpose
INFO	Normal operations
WARNING	Potential issues
ERROR	Application failures

Level	Purpose
CRITICAL	Major system failures

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6. Infrastructure Monitoring

Aegis Bank monitors infrastructure components.

Monitoring includes:

Server Health

Metrics:

- CPU usage.
- Memory usage.
- Disk utilization.
- Network activity.

Container Monitoring

Docker containers are monitored for:

- Availability.
- Resource usage.
- Failures.
- Restart events.

Cloud Monitoring

Cloud resources are monitored for:

- Performance.
- Security.
- Availability.

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7. Security Monitoring

Security monitoring detects suspicious activities.

Monitored events:

- Failed login attempts.
- Unauthorized access.
- Permission changes.
- Suspicious transactions.
- Security policy violations.

Security Monitoring Goals

- Detect attacks.
- Reduce response time.
- Maintain compliance.
- Protect customer data.

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8. Database Monitoring

Database systems are continuously monitored.

Metrics:

- Query performance.
- Connection activity.
- Storage usage.
- Failed access attempts.
- Backup status.

Database Alerts

Examples:

- Database unavailable.
- High resource usage.
- Unauthorized access attempt.
- Backup failure.

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9. API Monitoring

Aegis Bank monitors all critical APIs.

Tracked metrics:

Metric	Purpose
Response Time	Performance measurement
Error Rate	Failure detection
Availability	Service health
Traffic Volume	Usage analysis

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10. AI System Monitoring

Aegis AI OS requires specialized monitoring.

Monitored components:

- Document retrieval.
- Embedding generation.
- Vector search.
- LLM responses.
- Model performance.

AI monitoring tracks:

- Response quality.

- Retrieval accuracy.
- Processing time.
- Model behavior.

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11. LangFuse Integration

Aegis AI OS integrates LangFuse for AI observability.

LangFuse provides:

- LLM tracing.
- Prompt tracking.
- Response monitoring.
- Token usage analysis.
- Performance evaluation.

AI Request Flow Monitoring

Employee Query

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RAG Retrieval

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Document Selection

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LLM Processing

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Generated Answer

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LangFuse Trace

Example Trace Information

Field	Value
Query	Password Policy Request
Retrieved Source	AB-SEC-001
Model Used	GPT Enterprise Model
Response Time	2.3 seconds
Evaluation Score	94%

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12. Alert Management

Alerts notify teams about important events.

Alert Categories

Critical

Examples:

- Banking service outage.

- Security breach.
- Database failure.

Warning

Examples:

- High CPU usage.
- Increased errors.
- Slow response times.

Informational

Examples:

- Deployment completed.
- System updates.

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13. Incident Detection Process

When an issue occurs:

Monitoring Alert

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Incident Creation

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Team Notification

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Investigation

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Resolution

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Post-Incident Review

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14. Log Security Requirements

Logs must be protected.

Requirements:

- Access control.
- Encryption.
- Integrity protection.
- Audit tracking.

Sensitive information must not appear in logs:

❌ Passwords
❌ Encryption keys
❌ Full customer confidential data

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15. Retention & Storage

Monitoring data must follow retention requirements.

Stored information includes:

- Application logs.
- Security events.
- Audit records.
- AI traces.

Retention depends on:

- Regulatory requirements.
- Security needs.
- Business importance.

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16. Enterprise Monitoring Architecture

Aegis Bank uses a centralized monitoring architecture to maintain visibility across all banking systems.

Users

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Banking Applications

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API Services

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Monitoring & Logging Platform

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Metrics Logs Traces Alerts

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Operations + Security Teams

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17. SIEM Integration Standards

Aegis Bank integrates monitoring systems with Security Information and Event Management (SIEM) platforms.

The SIEM system collects and analyzes:

- Security events.
- Authentication activities.
- Network activities.
- Application logs.
- Database events.

SIEM Benefits

SIEM helps Aegis Bank to:

- Detect cyber threats.
- Investigate incidents.
- Monitor suspicious behavior.
- Maintain compliance records.

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18. Real-Time Security Monitoring

Security events are monitored continuously.

Examples:

Multiple Failed Login Attempts

Detection:

Multiple incorrect passwords

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Security Alert Generated

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User Activity Reviewed

Unauthorized Data Access

Detection:

Employee Access Attempt

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Permission Validation

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Access Approved / Blocked

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Audit Record Created

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19. AI Hallucination Monitoring

Aegis AI OS includes AI response quality monitoring.

The system evaluates:

- Answer accuracy.
- Source relevance.
- Retrieved documents.
- Confidence score.
- Hallucination risk.

AI Evaluation Workflow

Employee Question

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RAG Retrieval

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LLM Response

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Quality Evaluation

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Hallucination Score

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Final Answer

Example

Employee asks:

"Can employees approve loans above $50,000?"

AI checks:

1. Loan Approval Guidelines document.
2. Employee permissions.
3. Retrieved policy section.
4. Response accuracy.

Output:

Answer Confidence: 96%

Source:

AB-OPS-004 Loan Approval Guidelines

Section:

5.2 Approval Authority

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20. AI Performance Monitoring

Aegis AI OS tracks AI system performance.

Metrics include:

Metric	Purpose
Response Time	Measure speed
Retrieval Accuracy	Measure document matching
Token Usage	Track AI cost
Confidence Score	Measure reliability
Error Rate	Detect failures

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21. Aegis AI OS Observability Architecture

Employee Portal

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AI Gateway

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RAG Engine

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Vector Database

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LLM Services

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Evaluation Engine

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LangFuse

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Monitoring Dashboard

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22. Monitoring Dashboard Standards

Operations teams use dashboards to monitor system health.

Dashboards display:

Application Health

- Service availability.
- Errors.
- Response time.

Infrastructure Health

- Server status.
- Container status.
- Resource usage.

AI Health

- Query volume.
- Model performance.
- Retrieval quality.
- Failed responses.

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23. Incident Response Examples

Scenario 1 — Banking API Failure

Situation:

Customer payment API becomes unavailable.

Detection:

- Monitoring system detects increased errors.

Response:

1. Alert generated.

2. DevOps team notified.
3. Root cause analyzed.
4. Service restored.
5. Incident report created.

Scenario 2 — Security Threat Detection

Situation:

Multiple unauthorized login attempts detected.

Response:

1. SIEM generates alert.
2. Security team investigates.
3. User access reviewed.
4. Security controls activated.

Scenario 3 — AI Incorrect Response

Situation:

AI provides unsupported information.

Response:

1. LangFuse trace reviewed.
2. Retrieved documents checked.
3. Prompt evaluated.
4. Knowledge source updated.
5. Model performance improved.

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24. Monitoring Responsibilities

Site Reliability Engineering (SRE) Team

Responsible for:

- System monitoring.
- Alert management.
- Performance improvement.
- Availability management.

DevOps Team

Responsible for:

- Infrastructure monitoring.
- Deployment tracking.
- Container health.

Security Team

Responsible for:

- Threat detection.
- SIEM monitoring.
- Security investigation.

AI Engineering Team

Responsible for:

- Model monitoring.
- RAG evaluation.
- AI performance analysis.

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25. Monitoring Best Practices

Aegis Bank follows these practices:

Centralized Monitoring

All critical systems send monitoring data to approved platforms.

Automated Alerts

Critical issues generate automatic notifications.

Regular Reviews

Monitoring rules are reviewed regularly.

Data Protection

Monitoring information is protected like enterprise data.

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26. Frequently Asked Questions (FAQs)

Q1. Why is logging important for Aegis Bank?

Logging provides visibility, troubleshooting capability, and security tracking.

Q2. How does Aegis Bank detect security attacks?

Through continuous monitoring, SIEM integration, and security alerts.

Q3. How does Aegis AI OS monitor AI responses?

Through RAG evaluation, LangFuse tracing, confidence scoring, and source verification.

Q4. Can monitoring detect system failures automatically?

Yes. Automated monitoring detects failures and generates alerts.

Q5. Are customer passwords stored in logs?

No. Sensitive information must never be stored in application logs.

Q6. Why is LangFuse used?

LangFuse provides AI observability by tracking prompts, responses, latency, and model behavior.

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27. Related Documents

This document should be read with:

- **AB-IT-001 — API Development Standards**
- **AB-IT-002 — Database Standards**
- **AB-IT-003 — Docker Deployment Guide**
- **AB-IT-004 — CI/CD Standards**
- **AB-SEC-003 — Cyber Security Policy**
- **AB-COMP-003 — Risk Management Framework**

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28. Revision History

Version	Date	Description	Author
v1.0	March 20, 2026	Initial Logging & Monitoring Standards Release	SRE Team
v1.1	May 30, 2026	Added Security Monitoring and Alert Management	Infrastructure Team
v1.2	**July 24, 2026**	**Enhanced SIEM Integration, AI Observability, LangFuse Monitoring, and Enterprise Alerting Standards**	**Information Technology Department**

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29. Document Approval

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End of Document

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